

Homework 1

Overview

You will build a React application called Semester Course Tracker that displays your enrolled courses for the current semester and allows new courses to be registered.

Submission

Your work must be on GitHub and submitted via Google Classroom as a repository link. Create a new repository on GitHub, link it to your local project the way we did in the Git and GitHub session, and push your commits. The repository name must include your student ID number -- for example, if your ID is 211234, name it:

```
course-tracker-211234
```

Make sure your repository is set to Public so it can be accessed for grading.

Requirements

Create your project using Vite with the React template. Do not add any files or components yet.

Commit your changes with the message: `Created project`

Create a hardcoded array of your enrolled courses for this semester in `App.jsx`. Each course object must have these exact fields:

```
{
  id: number,
  name: string,
  credits: number,
  grade: number,
  attending: boolean,
  difficulty: string // 'Easy' | 'Moderate' | 'Hard'
}
```

The array must contain a minimum of three courses using your actual course names and credit hours. Then in `App.jsx`, display your full name, your student ID, and a short line indicating that the courses below are your enrolled courses for this semester. Below that, render the list of courses from your array.

Commit your changes with the message: `Listed courses and student info`

Create a `StudentCard` component that receives a course object as props and displays all of its fields. `StudentCard` must also show two conditional indicators based on the course data, using exactly these conditions:

```
// At risk
```

```
grade < 50 && attending === false  
  
// Eligible for distinction  
grade >= 85
```

Commit your changes with the message: Added StudentCard

Create a RegisterCourse component that allows a new course to be added to the list. The form must include five inputs in this order: course name (text), credit hours (number), current grade (number), attending regularly (checkbox), and difficulty (select with options Easy, Moderate, Hard), followed by a Register button. The form must validate that the course name is not empty and that the grade is a number from 5 to 10.

Commit your changes with the message: Added RegisterCourse

Add a small but meaningful use of useRef and useCallback to your application. Include one inline comment somewhere in your code that explains a non-trivial decision in your own words. A comment that only describes what a line does is not acceptable.

Commit your changes with the message: Added additional functionality

Summary

When the assignment is complete, your project must have two components -- StudentCard and RegisterCourse -- both in the components folder. App.jsx renders your student information, the list of courses using StudentCard, and the RegisterCourse form. All five commits must be present with the exact messages specified above.